



Tribhuvan University
Institute of Science and Technology



Himalaya College of Engineering
Chyasal, Lalitpur



Project proposal

TutorLink – A Web-Based Home Tutition Platform

Submitted to:

Department of CSIT

Himalaya College of Engineering, Chyasal, Lalitpur

*In Partial Fulfilment of the requirements for the degree of Bachelor of Science,
Computer Science and Information Technology(BSc. CSIT)*

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1. Introduction

1.1 Introduction

In today's competitive academic environment, students often require personalized guidance and focused attention beyond what is available in traditional classrooms. As a result, the demand for home tuition has increased significantly among parents and students who seek customized learning support to improve academic performance, strengthen weak subjects, and prepare effectively for examinations.

Despite this growing demand, the existing home tuition ecosystem operates largely through informal networks, word-of-mouth references, social media posts, and brokerage agents. Brokers or middlemen typically act as intermediaries between tutors and parents, charging substantial commissions from tutors before arranging tuition agreements. This practice not only reduces the income of tutors but also creates issues of trust, transparency, and reliability for both parties. Parents often struggle to verify the qualifications and background of tutors, while tutors face financial exploitation and uncertainty regarding genuine tuition opportunities.

With the rapid advancement of web technologies and digital platforms, it is now possible to replace these inefficient and unstructured processes with a centralized, secure, and transparent system. A web-based platform can provide verified tutor profiles, efficient search mechanisms, secure communication, and fair interaction without the involvement of brokers. Such a system can significantly reduce the time, cost, and effort required to arrange home tuition while building trust between students, parents, and tutors.

TutorLink is proposed as a web-based home tuition facilitation platform designed to address these challenges. The system directly connects students and parents with qualified tutors based on subject expertise and geographical location. Tutors can create detailed profiles showcasing their qualifications and teaching experience, while students and parents can search and select tutors according to their requirements. The platform also incorporates secure authentication, controlled contact requests, and administrative monitoring to ensure reliability and data integrity.

By eliminating brokerage dependency and introducing a transparent digital environment, TutorLink aims to modernize the traditional home tuition system and make it more accessible, fair, and trustworthy. This project demonstrates how modern web technologies can be applied to solve real-world problems in the education sector, ultimately contributing to a more efficient and student-centered learning support system.

1.2 Problem Statement

The existing home tuition system suffers from several problems:

- Excessive brokerage charges imposed on tutors
- Lack of transparency in agreements between tutors and parents
- Difficulty in finding suitable tutors or tuition opportunities
- Absence of a centralized and trustworthy platform

- No proper verification of tutor qualifications
- Time-consuming and unreliable manual processes

Due to these issues, both tutors and parents face inconvenience, financial loss, and trust-related problems. There is a strong need for a system that eliminates these inefficiencies and provides a direct, fair, and transparent environment for home tuition services.

1.3 Objectives of the Project

General Objective

To develop a web-based platform that directly connects students and tutors, eliminating the need for brokers and ensuring transparency, trust, and convenience.

Specific Objectives

- To provide a reliable platform for students and parents to find home tutors
- To help tutors find genuine home tuition opportunities without brokerage
- To enable subject-wise and location-based tutor search
- To ensure secure user authentication and data handling
- To create verified tutor profiles for informed decision-making
- To reduce time, cost, and effort involved in traditional home tuition arrangements

1.4 Scope and Limitations

Scope

TutorLink focuses on:

- Tutor registration with detailed profiles (qualification, subjects, location)
- Student/parent ability to search tutors based on subject and location
- Secure contact request system between students and tutors
- Admin panel for user and system management
- Transparent and broker-free tuition facilitation

Limitations

- The system is limited to a web-based platform only
- No live online classes or video conferencing in current version
- No mobile application support in the initial version

1.5 Report Organization

- **Chapter 1:** Introduction, problem statement, objectives, and scope of the project.
- **Chapter 2:** Background study and literature review of existing systems and research gap.

- **Chapter 3:** System analysis including existing system problems, proposed system, requirements, and feasibility.

2. Background Study and Literature Review

2.1 Background Study

Home tuition services traditionally operate through personal networks, word-of-mouth, social media groups, and brokerage agencies. These approaches are informal and lack proper verification, documentation, and accountability.

General concepts related to this project include:

- Web-based service platforms
- User authentication and profile management
- Search and filtering systems
- Role-based access control
- Secure data handling and communication

These concepts form the foundation for developing a reliable home tuition facilitation system like TutorLink.

2.2 Literature Review

Many tutoring and education platforms have emerged to facilitate learning by connecting students with tutors through online systems. These platforms vary in structure, services, and effectiveness, and have both strengths and limitations.

Existing commercial tutoring marketplaces such as **Wyzant** allow students to filter tutors by subject, experience, and ratings, providing flexibility and choice for users. Wyzant's detailed tutor profiles and pay-as-you-go model increase transparency and enable personalised learning, but the platform also adds service fees and has variable tutor quality due to independent contractors [1].

Platforms like **Tutor.com** and **Chegg Tutors** have broad subject coverage and long operational histories, offering on-demand help across academic levels and accessible user interfaces. Chegg often includes interactive tools and affordable pricing; however, research reports potential issues such as varying tutor expertise and technical disruptions [2].

Preply provides a global marketplace with AI-enhanced tutor matching and integrated classroom tools such as video and chat, helping learners find specialised tutors worldwide. However, commission costs and high competition among tutors can decrease tutor earnings [3].

Other platforms, such as **Tutoria**, include strict tutor verification processes and a wide range of subjects, which help improve trust and quality. Yet, they often charge high commission rates per booking, limiting tutor income potential [4].

Research indicates that digital tutoring systems contribute to improved learning outcomes, flexibility in scheduling, and personalised instruction compared to traditional classroom settings. The adoption of online platforms has been shown to correlate with student satisfaction and effectiveness in learning, highlighting the growing relevance of technology-mediated education [5].

Despite these benefits, literature also shows notable challenges. Technical issues, unstable internet connectivity, and inconsistent tutor quality can negatively impact learning experiences on online platforms, emphasizing the need for robust design and quality control mechanisms [6].

A systematic review of tutoring systems further highlights trends in educational technology research, such as the integration of adaptive and AI-based tutoring models. It also notes concerns about scalability, ethical considerations, and variability in instructional quality [7].

In summary, existing systems demonstrate benefits such as accessibility, personalised matching, and flexible scheduling, but they also present drawbacks like high fees, inconsistent quality control, technical issues, and limited localisation for home tuition markets. These insights highlight the research gap that **TutorLink** aims to address by developing a **broker-free, locally relevant, transparent, and quality-focused home tuition platform**.

2.3 Research Gap

There is a clear research gap in developing a system that:

- Eliminates brokerage dependency
- Provides localized support for home tuition
- Ensures transparency and trust between tutors and parents
- Offers verified profiles and secure communication

TutorLink is proposed to fill this gap by addressing the specific problems present in the traditional home tuition ecosystem.

3. System Analysis

3.1 Requirement Analysis

Functional Requirements

FR ID	Functional Requirement	Description
FR-01	User Registration	The system shall allow students/parents and tutors to register by providing required personal details.

FR-02	User Login/Authentication	The system shall authenticate users securely using username/email and password.
FR-03	Tutor Profile Management	The system shall allow tutors to create, update, and manage profiles including qualification, subjects, and location.
FR-04	Search Tutors	The system shall allow students/parents to search tutors based on subject and location.
FR-05	View Tutor Details	The system shall display detailed tutor profiles to students/parents.
FR-06	Contact Request System	The system shall allow students/parents to send contact requests to tutors.
FR-07	Payment Control for Contact Access	The system shall restrict tutor contact details until successful payment is completed.
FR-08	Notification System	The system shall send notifications for requests, approvals, and updates to users.
FR-09	Admin Login	The system shall allow admin to log in with privileged access.
FR-10	User Management by Admin	The admin shall be able to view, manage, and remove users if necessary.
FR-11	Gig/Tuition Request Management	Students shall be able to post tuition requirements and tutors shall be able to apply.
FR-12	Premium Feature for Tutors	The system shall allow tutors to activate premium visibility features.
FR-13	Gig Completion Status	Tutors and students shall be able to mark tuition gigs as completed.

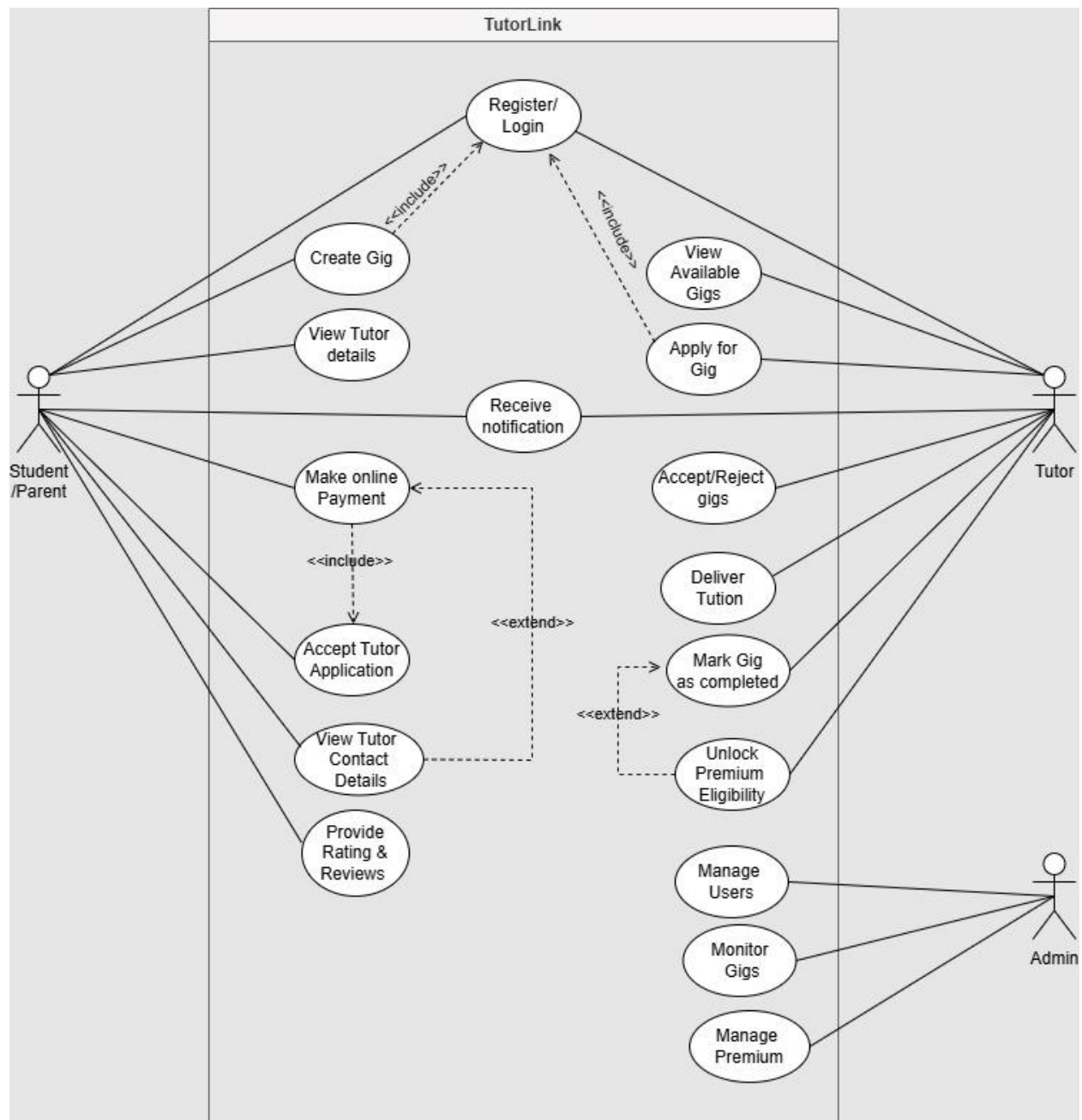


Fig:Use Case Diagram

Non-Functional Requirements

NFR ID	Non-Functional Requirement	Description
NFR-01	Security	The system shall encrypt passwords and protect user data using authentication and authorization mechanisms.

NFR-02	Performance	The system shall provide fast response time during tutor search and page navigation.
NFR-03	Usability	The system shall provide a simple, user-friendly interface for easy navigation.
NFR-04	Reliability	The system shall operate consistently without failure during normal usage.
NFR-05	Availability	The system shall be accessible to users at all times except during maintenance.
NFR-06	Scalability	The system shall be able to handle increasing numbers of users and data.
NFR-07	Maintainability	The system shall be easy to update and maintain by developers.
NFR-08	Compatibility	The system shall work properly on major web browsers (Chrome, Edge, Firefox).
NFR-09	Data Integrity	The system shall ensure accurate storage and retrieval of user and gig data.
NFR-10	Privacy	The system shall restrict unauthorized access to personal user information.

3.2 Feasibility Analysis

Technical Feasibility

The development of TutorLink is technically feasible due to the availability of modern, reliable, and open-source web technologies. The system is designed using **React JS** for the frontend, **Django REST Framework** for the backend, and a **relational database** for data management. These technologies are widely used, well-documented, and supported by large developer communities, ensuring stability and scalability.

The development tools, frameworks, and libraries required for implementation are freely available and compatible with standard development environments. Integration with online payment gateways such as **Khalti** is also technically achievable through REST APIs. Since the project team possesses the required knowledge and skills in these technologies, the system can be developed efficiently without technical barriers.

Furthermore, the system architecture supports modular development, making it easier to test, maintain, and enhance in the future. Therefore, from a technical perspective, TutorLink is fully feasible and implementable.

Economic Feasibility

TutorLink is economically feasible because it relies entirely on **open-source technologies**, eliminating the need for expensive software licenses. The primary costs involved are limited to development time, internet resources, and minimal hosting expenses.

Unlike traditional home tuition systems where tutors lose a significant portion of their income to brokers, TutorLink reduces financial exploitation by providing a direct connection between tutors and students. This creates a cost-effective solution for users while also ensuring sustainability of the platform.

In the long term, the system can generate revenue through optional premium features for tutors and minimal service charges for contact access, making it financially sustainable without imposing heavy costs on users.

Operational Feasibility

The system is designed with a strong focus on **usability and user convenience**. The interface is simple, intuitive, and easy to navigate, allowing students, parents, and tutors to use the platform without requiring technical expertise or training.

Since the platform automates the traditional manual process of finding tutors and arranging tuition, users can save significant time and effort. The admin panel ensures smooth monitoring and management of system activities, maintaining operational efficiency.

Given the increasing familiarity of users with web applications and online services, TutorLink can be easily adopted in real-world scenarios. Hence, the system is operationally feasible and practical for everyday use.

Social Feasibility

TutorLink contributes positively to society by promoting **fairness, transparency, and trust** in the home tuition sector. It reduces dependency on brokers, prevents financial exploitation of tutors, and helps parents find verified tutors easily.

By digitizing the tuition arrangement process, the system supports modernization in the education sector and encourages the use of technology for social benefit.

3.3 Analysis Approach

The system follows an **Object-Oriented Approach (OOA)** for analysis and design.

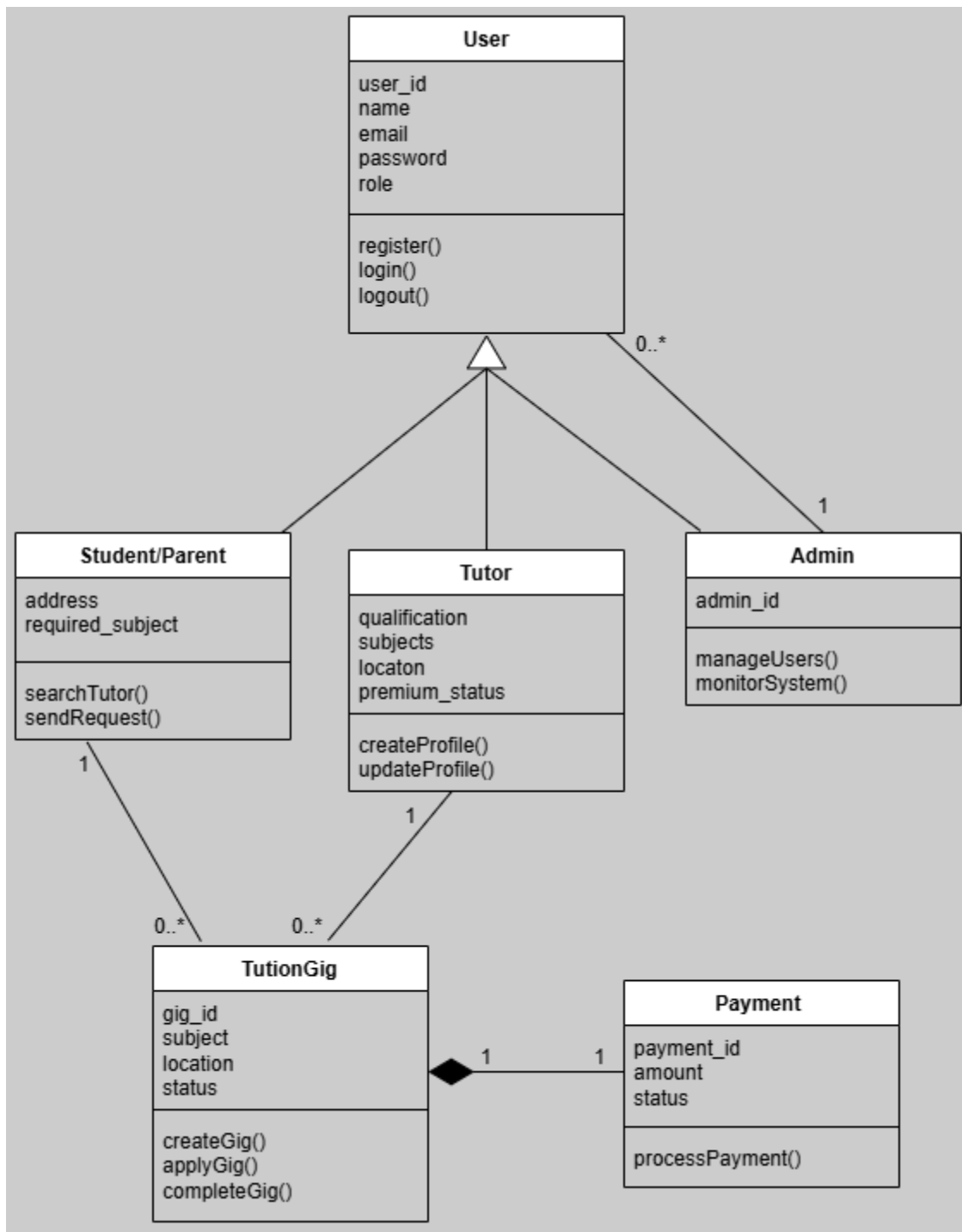


Fig: Class Diagram

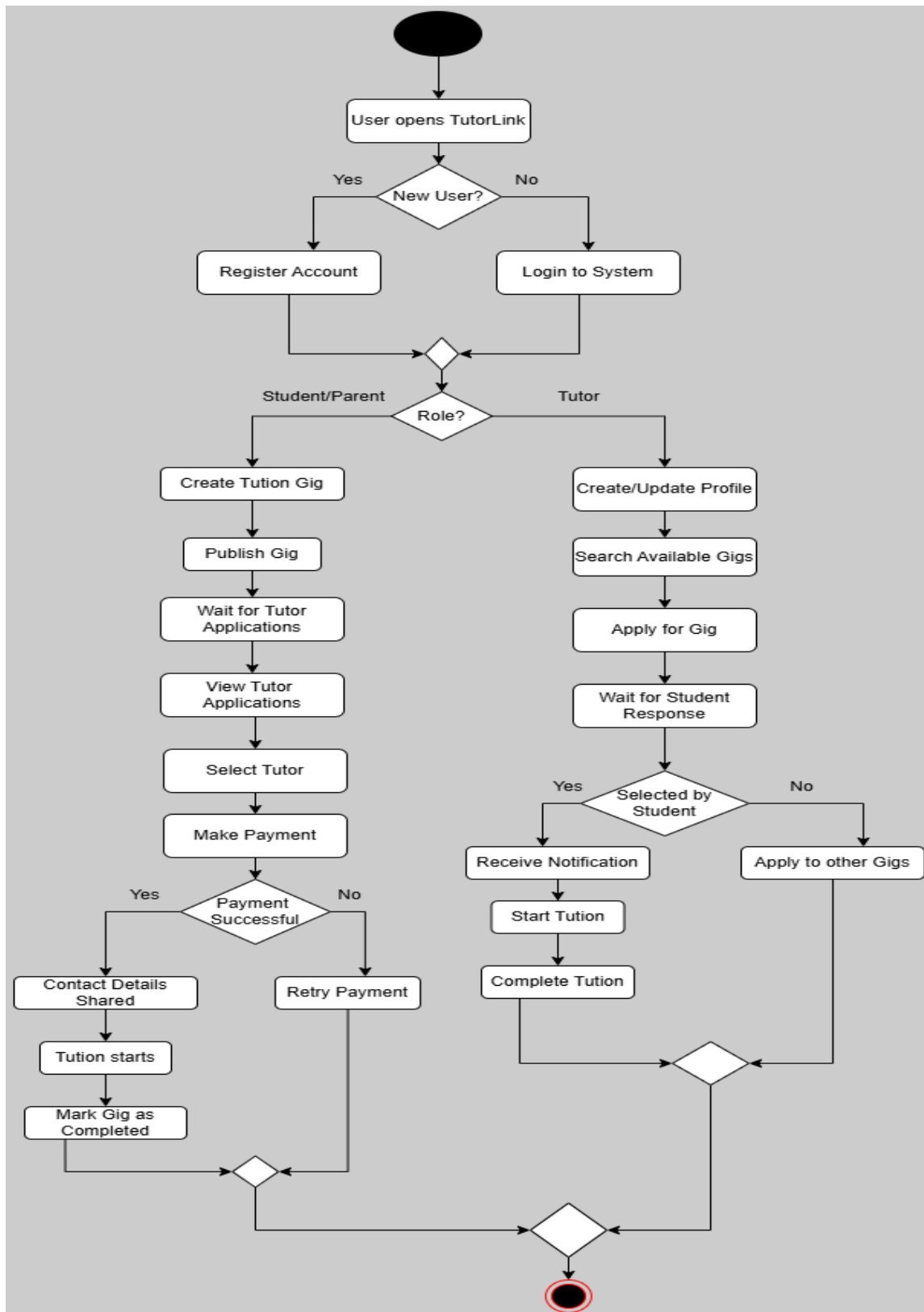


Fig: Activity Diagram

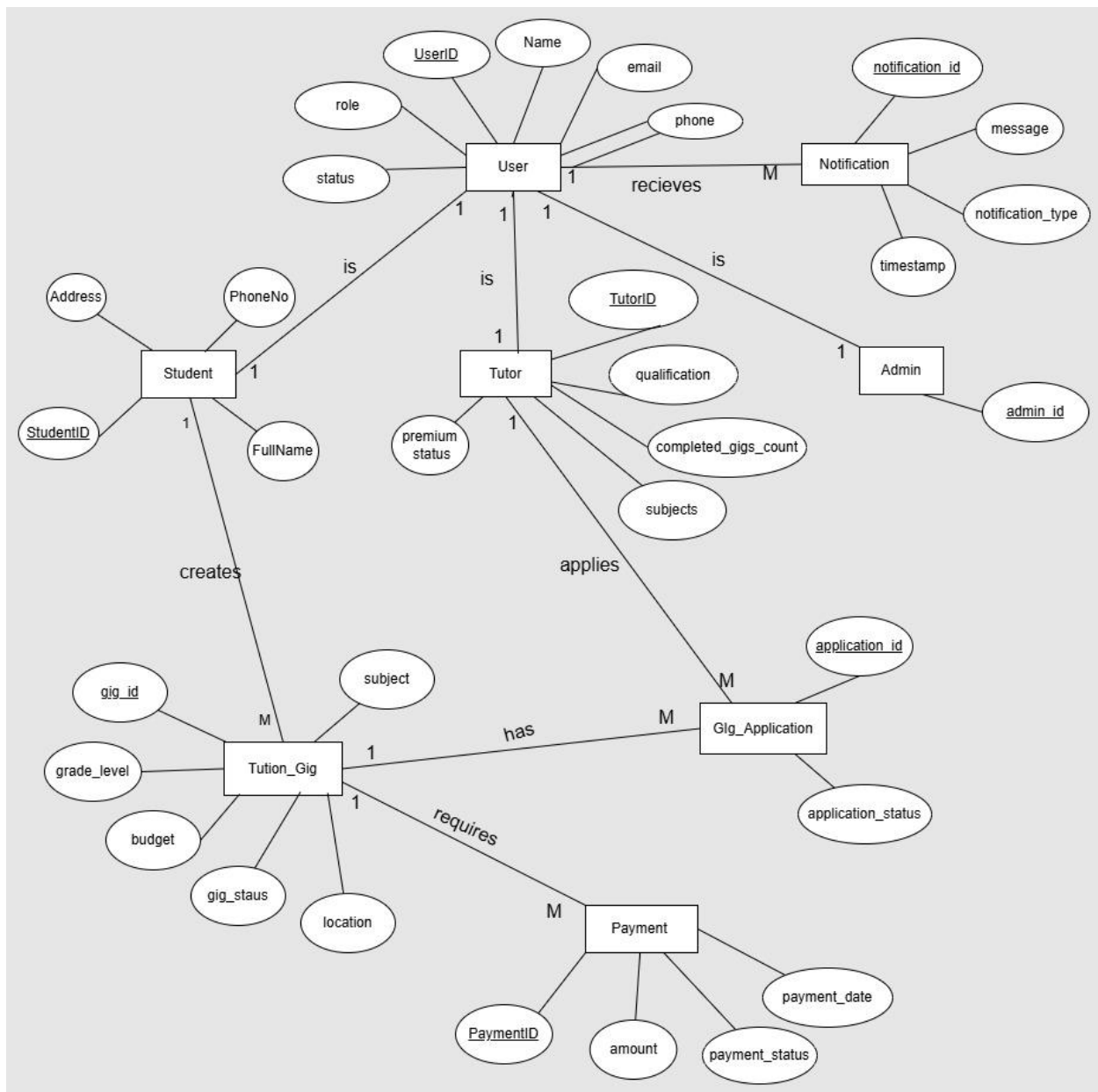


Fig: Entity-Relationship Diagram

References

- [1] “Top 10 Tutoring Marketplace Platforms – Features, Pros & Cons,” Cotocus, 2025.
<https://www.cotocus.com/blog/top-10-tutoring-marketplace-platforms-features-pros-cons-comparison/>
- [2] “Top Online Tutoring Platforms – Compared,” Education.Antaranews, 2025.
<https://education.antaranews.com/top-online-tutoring-platforms-compared-pros-cons-pricing/>
- [3] “Best Tutoring Software,” TutorCruncher, 2025. <https://tutorcruncher.com/blog/best-tutoring-software/>
- [4] “Tutera,” Wikipedia, 2025 . <https://en.wikipedia.org/wiki/Tutera>
- [5] A. Author, “Investigating the Effectiveness and Student Satisfaction of Online Learning Platforms in Comparison to Traditional Classroom Settings,” *GSSR Journal*, vol. X, pp. XX–XX, 2025. <https://www.gssrjournal.com/article/investigate-the-effectiveness-and-student-satisfaction-of-online-learning-platforms-in-comparison-to-traditional-classroom-settings>
- [6] “Online Tutoring Platforms Overview – Pros & Cons,” BrightDomo, 2025.
<https://brightdomo.com/online-tutoring-platforms-overview>
- [7] J. Doe, R. Smith, and L. Sanchez, “Educational Technology and Tutoring Systems: Trends, Challenges, and Future Research,” *arXiv preprint arXiv:2503.09748*, 2025.
<https://arxiv.org/abs/2503.09748>